2026 Basic Abstract Algebra Study

LECTURE 2 HOMEWORKS

SOLUTION - Q2

Version
v1.0.0

Instructor
Abiria

Published on
2026-05-30

Hosted by
Coding Lab

Note 1 (Notations)

The following notations are used throughout the questions below.

- $\mathbb{Z}$: The set of all integers.
- $\mathbb{Z}^+$: The set of all positive integers; i.e., $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$.
- $\mathbb{Z}_n$: The set of non-negative integers from 0 to $n - 1$ for some positive integer n. For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $\mathbb{R}$: The set of all real numbers.
- $\min(x, y)$: The minimum of x and y; i.e., the smaller of the two.
- $\mathcal{P}(X)$: The power set of the set X.
- μ_n: The set of all n-th roots of unity.

Q1.

EASY

Let $f, g, h : \mathbb{Z} \to \mathbb{Z}$ be given by

$$f(x) = x + 3, \quad g(x) = 2x, \quad h(x) = x - 1.$$

What is the value of $(h \circ g \circ f)(1)$?

○ **1.** 3 ○ **2.** 5 ○ **3.** 7 ○ **4.** 8 ○ **5.** 9

Q2.

EASY

Let S_5 be the set of all bijections from $\mathbb{Z}_5$ to $\mathbb{Z}_5$.

What is the value of $|S_5|$?

○ **1.** 5 ○ **2.** 25 ● **3.** 120 ○ **4.** 125 ○ **5.** 625

As covered in the lecture, the number of bijections from a finite set of size n to another set of size n is given as $n!$.

A quick calculation shows that $|S_5| = 5! = 2 \cdot 3 \cdot 4 \cdot 5 = 120$.

Therefore, the third choice is the correct answer.

Definition 2 (Injection)

A function $f : X \to Y$ is said to be *injective*, or called an *injection*, if for every $x, y \in X$ with $x \neq y$, we have $f(x) \neq f(y)$.

Q3.

EASY

Classify each of the following functions as injective or NOT injective. Which one differs from the rest?

○ **1.** $f_1 : \mathbb{R} \to \mathbb{R}$, $f_1(x) = \min(x, 2x)$ ○ **2.** $f_2 : \mathbb{R} \to \mathbb{R}$, $f_2(x) = 0 \cdot x$

- 3. $f_3 : \mathbb{Z} \to \mathbb{Z}$, $f_3(x) = \gcd(x, 10)$
- 4. $f_4 : \mathbb{R} \to \mathbb{R}$, $f_4(x) = x^2 + 5x + 6$
- 5. $f_5 : \mathbb{Z} \to \mathbb{Z}$, $f_5(x) = |x|!$

Definition 3 (Equality of functions)

Two functions with the same domain and same codomain are said to be *equal* if and only if they map every element of the domain to the same element of the codomain.

In other words, if $f, g : X \to Y$ are functions with the same domain and codomain, then $f = g$ if and only if $f(x) = g(x)$ for every $x \in X$.

Q4.

EASY

Given that every function below has $\mathbb{Z}^+$ as both its domain and codomain, which one differs from the rest?

- 1. $f_1(x) = \gcd(x, x^2) + \gcd(x, x^3)$
- 2. $f_2(x) = \operatorname{lcm}(2x, 3x) - \operatorname{lcm}(x, 3x)$
- 3. $f_3(x) = {}_{x+1}C_x + x - 1$
- 4. $f_4(x) = \gcd(2x, 6x)$
- 5. $f_5(x) = |3x| - |x|$

Definition 4 (Idempotent function)

A function f from a set X to itself is called *idempotent* when $f \circ f = f$.

Q5.

EASY

Which of the following functions is NOT idempotent?

- 1. $f_1 : \mathbb{Z} \to \mathbb{Z}$, $f_1(x) = x \bmod 5$
- 2. $f_2 : \mathbb{R} \to \mathbb{R}$, $f_2(x) = \min\left(\sqrt{2}, x\right)$
- 3. $f_3 : \mu_{64} \to \mu_{64}$, $f_3(x) = x^8$
- 4. $f_4 : \mathbb{R} \to \mathbb{R}$, $f_4(x) = |2x| - |x|$
- 5. $f_5 : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $f_5(x) = x \cup \mathbb{Z}$

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark × in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark ○ in the Evaluation column.

No.	Evaluation	No.	Evaluation
Q1		Q4	
Q2		Q5	
Q3			

There are also a few additional questions; feel free to answer them.

(a) Which was the hardest for you, among those you have solved? __________

(b) Which seemed to be the hardest for you, among those you haven't solved? __________

(c) Which was your favorite question, if you had to choose one? __________